

Hengyi Li

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EDUCATION



University of Amsterdam

MSc. Mathematics (Stochastics)

Sep. 2023 – Aug. 2026

- **Selected Coursework:** Measure-Theoretic Probability, Stochastic Processes, Stochastic Simulation, Asymptotic Statistics, Forensic Probability and Statistics, Statistical Mechanics, Mathematical Biology, Data-Driven Decision Making in Operations Research.



University of Liverpool

Xi'an Jiaotong-Liverpool University

BSc. Applied Mathematics

Sep. 2019 – Jul. 2023

- **Selected Coursework:** Complex Functions, Metric Spaces, Functional Analysis, Ordinary Differential Equations and Control, Partial Differential Equations, Numerical Analysis, Advanced Linear Algebra, Optimisation Theory.

Daqing Experimental Middle School

High School Diploma

Sep. 2016 – Jun. 2019

RESEARCH

Accelerating Convergence to Equilibrium in Gaussian Diffusions

Master Thesis, supervised by Dr. S.G.Cox [\[Latest\]](#) [\[Repository\]](#)

Mar. 2026 – Aug. 2026

- Characterised linear diffusions preserving a prescribed Gaussian invariant measure via the Lyapunov equation and analysed convergence in L^2 -Wasserstein distance through spectral properties of the drift matrix.
- Derived and compared optimal reversible and non-reversible convergence rates under noise constraints, including explicit low-dimensional constructions and the arbitrary-dimensional optimisation problem.
- Applied the framework to kinetic Langevin dynamics and examined extensions to strongly convex non-Gaussian potentials.

Random Walk and Boundary Value Problems

Bachelor Thesis, supervised by Dr. Ye Liu [\[PDF\]](#)

Sep. 2022 – May. 2023

- Constructed simple random walk models across multiple dimensions to analyse probability distributions and recurrent/transient properties, and applied these models to successfully tackle boundary value problems.

SELECTED ACADEMIC PROJECTS

Superinfections Can Induce Evolutionarily Stable Coexistence of Pathogens

Course Essay, Mathematical Biology [\[PDF\]](#)

Nov. 2025 – Jan. 2026

- Analysed epidemiological models incorporating superinfection to combine two scales of natural selection, demonstrating that superinfection can imply higher virulence and yield an evolutionarily stable coexistence.

Lee–Yang Theorem: Foundations, Proof and Applications

Course Essay, Statistical Mechanics, supervised by Prof. dr. W.M.Ruszel [\[PDF\]](#)

Dec. 2024 – Jan. 2025

- Presented theoretical underpinnings from complex analysis for the Lee–Yang theorem, demonstrating that the Ising model partition function remains non-vanishing within specific complex domains to guarantee analytic continuation of infinite-volume pressure and ensure analyticity of connected correlation functions.

Evaluating String Theory Through the Lens of Scientific Realism

Course Essay, Philosophy of Physics, written with Omar Shah [\[PDF\]](#)

Nov. 2024 – Dec. 2024

- Analysed the contemporary debate surrounding string theory and applied core arguments for and against scientific realism to address criticisms of the theory's unifying framework.

Contemporary Undergraduate Mathematical Contest in Modelling

Team Modelling Project

Sep. 2019

- Used numerical simulation to construct differential models and optimisation models for the inside pressure of the high-pressure oil pipe, and made a control scheme for the pump and pressure-reducing valve.

INTERNSHIP EXPERIENCE

Ping An Insurance Company of China

Analyst, the Car Insurance Department

Jul. 2022

- Updated the historical data of auto insurance projects and calculated the variance, covariance, and probability distribution of the data to find out VaR at 95%.
- Constructed risk factor regression models to analyse the impact of the auto-related indicators, and used the Monte Carlo simulation to confirm and adjust the models based on historical data.

Daqing High-tech City Construction Investment Department Company

Intern, the Department of Construction and Economy

Jun. 2022

- Participated in the smart city project and learned the application of machine learning models in spatial statistics.
- Used TensorFlow to label space vector features and used SVM models to tag and mine features.

Kaiyuan Securities

Financial Analyst, the Securities Investment Advisory Sector

Jul. 2020 – Aug. 2020

- Attended business training to understand the process of securities trading, including account opening, commissioning, fund transfer, trading, settlement, loss, account cancellation, etc., and carry out simulated practical operations.
- Used Python to perform statistical analysis on major financial assets and constructed probability models to make predictions and provide data support for the company's asset management.

SKILLS

Languages: **Chinese** (native speaker), **English** (language of instruction), **Japanese** (fluent).

Skills: Java, Python, MATLAB, hugin, Microsoft Excel, Adobe After Effects, etc.

Interests: Literature, philosophy, history, and visual works, including paintings, movies, and experimental videos.